

Innam Ul Haq

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PROFILE

AI-focused software engineer building LLM-powered applications, RAG pipelines, and agentic workflows using Python and FastAPI. Experienced in developing scalable AI systems integrating vector retrieval, API orchestration, and real-time AI workflows.

EDUCATION

Sharda University <i>Bachelor of Science in Computer Science (Artificial Intelligence & Machine Learning)</i> • CGPA: 8.89/10.0 - Demonstrating consistent academic excellence	Greater Noida, Uttar Pradesh 2026
BPS Baramulla <i>12th Standard</i>	Baramulla, Jammu And Kashmir Completed 2020
BPS Baramulla <i>10th Standard</i>	Baramulla, Jammu And Kashmir Completed 2018

TECHNICAL SKILLS

Programming Languages: Python, SQL, Java
Backend & APIs: FastAPI, REST APIs, API Integration, JWT Authentication, Webhooks, Async Workflows, Next.js
Generative AI & LLM Systems: RAG Pipelines, Prompt Engineering, Tool Calling, Agentic Workflows, Conversational AI, LLM Orchestration, OpenAI APIs, Groq APIs, Hugging Face
Vector Databases & Data: ChromaDB, SQLite, Pandas, NumPy
AI/ML Frameworks: PyTorch, Scikit-learn, Transformers
Tools & Platforms: Git/GitHub, Docker (Learning), VS Code, Cursor, Jupyter Notebook

PROJECTS

- AI Powered Investment Intelligence Platform** | *Python, FastAPI, Next.js, Groq, ChromaDB, RAG, JWT*
- Architected a full-stack multi-tenant investment research platform where analysts query any company in natural language and receive real-time structured analysis: stock metrics, sentiment-classified news, and AI-generated risk assessments.
 - Built an agentic LLM pipeline using **Groq (Llama 3.3 70B)** with dynamic tool calling across three specialized tools: live stock data (Yahoo Finance), news sentiment analysis (NewsAPI), and **RAG based retrieval** over financial documents (ChromaDB vector store).
 - Engineered a secure backend with FastAPI featuring **JWT authentication**, organisation level multi tenancy, and role based access control.
 - Designed a responsive Next.js dashboard rendering query results as structured cards, interactive stock price charts, sentiment badges, and source attributed insights.
 - GitHub Repository**
- End to End AI Voice Agent for Real Estate Customer Engagement** | *Python, FastAPI, OpenAI, ElevenLabs*
- Developed an end to end conversational AI voice agent capable of interacting with customers, sharing construction updates, and assisting with site visit scheduling through natural voice conversations.
 - Orchestrated a speech to speech AI pipeline using OpenAI Whisper (STT), GPT-4o mini (LLM), and ElevenLabs (TTS) with a FastAPI backend and browser based MediaRecorder interface for real time voice interaction
 - Implemented a browser based interface using HTML, CSS, and JavaScript with the MediaRecorder API to capture real time voice input and enable seamless voice interaction.
 - Designed a scalable queue-based architecture to support automated AI outreach to 1000+ customers simultaneously.
 - GitHub Repository**
- Multimodal Hateful Meme Classification using Iterative Cross-Attention** | *Python, PyTorch, BERT, CLIP, NLP*
- Designed and implemented a deep multimodal classification framework to detect hateful memes by jointly reasoning over image and text modalities.
 - Proposed an **iterative cross-attention fusion architecture** integrating frozen CLIP visual embeddings with a fine-tuned BERT text encoder across all transformer layers.

- Addressed semantic incongruity in memes through layer wise visual grounding, enabling context-aware hate speech detection.
- Trained and evaluated the model on the Facebook Hateful Memes Challenge dataset, achieving **67.18% Macro F1 score** and **67.20% accuracy**.
- Implemented class imbalance mitigation using weighted loss, weighted sampling, cosine learning rate scheduling, and early stopping.
- **Conference Paper Presentation** Received certificate for presenting this research at an academic conference (ICAIM).

CERTIFICATIONS

Python for Everyone: Master The Basics of Programming

Sayman Creative Institute

- * Comprehensive certification program covering Python fundamentals essential for data science applications
- * Gained extensive hands on experience in Python programming including loops, functions, data types, and object oriented programming concepts
- * Developed strong problem-solving skills through practical coding exercises and real world programming challenges
- * Built solid foundation in Python programming that directly supports machine learning and data analysis projects
- * Certificate Link: [View Certificate](#)

Certificate of Paper Presentation — ICAIM-2025

International Conference Presentation

- * **ICAIM**: International Conference on Applied and Interdisciplinary Mathematical Sciences (ICAIM-2025).
- * Received a certificate for presenting the research paper “**Solving Semantic Incongruity: An Iterative Cross-Attention Framework for Robust Hateful Meme Classification**”.
- * Presented work focused on multimodal deep learning using CLIP + BERT with iterative cross-attention fusion for hateful meme classification.
- * [View Certificate](#)